

# ***Pocket Therapist: An AI-Powered Journaling and Mental Wellness Companion***

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**Abstract:** Pocket Therapist supports stress management and healthy habits through low-effort journaling, combining text and voice input with on-device emotion analysis and passive activity tracking.

**Keywords:** mobile wellness, journaling, emotion analysis, step tracking, user centered design

## **1. Introduction**

### **1.1 Recap of the Problem**

Young adults are experiencing higher levels of stress, anxiety and burnout than ever before, while access to long-term mental health support remains elusive. The act of journaling is an established means of increasing emotional insight and developing resilience; however, most digital journaling applications require significant input at times of high stress and do not consider entries in conjunction with other data sources. Consequently, many of these applications ignore critical aspects of one's life such as, work routines, physical activity, and other contextual factors resulting in poor advice, unreliability, and low usage frequency.

### **1.2 Who Is the Audience?**

This is designed specifically for young adults who use mobile devices to keep track of their responsibilities and maintain their health. Due to their busy and ever-changing lifestyles, plus their tendency to experience high levels of stress, many of these individuals will find any intrusive or demanding tools very problematic. Because of this, there will be a very high level of sensitivity towards any tools that may be used to collect information about their mental health status. To overcome this sensitivity, the solution must be low-effort, mobile first, privacy aware, and aware of the current conditions of the user when they use the tool.

### **1.3 High-Level Description of the Solution**

Pocket Therapist is a context-aware, AI-driven agent for mental health that makes it easy for the user to perform reflective journaling with the least amount of cognitive load. The service does not work like a conventional journaling application but rather provides a supportive tool that blends voice input, mobile sensing, and on-device intelligence to derive emotional state information from the user and provide him/her with personalized assistance based upon this data. By decreasing the friction of interactions between the user and mobile mental health support, while grounding the user's insights

within signals from their environment, Pocket Therapist provides better relevance of suggestions and insights while maintaining privacy and autonomy through explicit permissions. In the following sections the design choices are discussed as they relate to a context-aware, privacy-conscious approach to delivering mobile mental health support.

## **2. Background and Related Work**

### **2.1 Necessary Background**

Therapeutic journaling promotes awareness of emotions and the ability to manage stress through helping individuals recognize the context of their thoughts. However, effectiveness can vary based on when, how and how frequently the journaling process is performed. The use of mobile devices to determine the emotional state of a user (such as sentiment detection) and the user's "activity" or daily routine using sensors is becoming widely adopted in healthcare, however most current implementations only utilize one of these methods at a time. Therefore, building a combined mobile system that provides therapeutic journaling, sentiment detection and sensor data into one Privacy Respected System is a Critical Issue for many and is one of the primary driving factors for building the Pocket Therapist platform.

### **2.2 Existing Applications and Research**

Most commercial mental wellness applications provide content that is either pre-recorded (including voice and music) or relies on users' own reflections, such as guided meditations (Calm and Headspace) that are delivered via recorded audio experiences. While these services encourage mindfulness, they provide few personalization options and have simplified user experience to make them easy to use. Other applications (Daylio, Apple Journal, and Wysa) focus solely on tracking mood or keeping a journal, providing a way for users to track their mood through analysis or respond to a conversation. Although all these applications increase users' awareness of their emotional state, they rely solely on the user's ability to input information (display their mood) and typically don't employ any context-based input. Social scientific and computer science prototypes of emotion inference using language and detection of mobile sensor-generated activity via sensors have been tested; however, most studies evaluate emotion inference and mobile sensor-generated activity technology used in conjunction with end-to-end wellness applications that a user will encounter with respect to mobile issues, including consuming battery, doing a private meditation, and having a real-world, usable experience.

### **2.3 Shortcomings of Prior Solutions**

Fragmentation is a core issue for all approaches: commercial and research alike. Journaling, emotional inference, and contextual awareness all exist as separate products, creating disconnection between these two processes and how they support each other;

furthermore, the reliance on manual interaction during high-stress situations significantly detracts from user commitment to using any of the tools offered by the providers. In addition, AI recommendations are often vague, and there is little agency over how personal information is collected and stored by providers due to the lack of communication regarding these aspects of service usage. The deficiencies of these products compel a holistic, systemic solution to journaling, emotional inference, and contextual awareness with an emphasis on low effort, explainable outcomes, and user agency. Pocket Therapist will fulfil these needs via on-device machine learning, sensor fusion, and design that is privacy-centric.

### 3. High-Level Overview of the Solution

The Pocket Therapist's Architecture is an all-inclusive framework that continually interconnects user expression, sensing in context, intelligent inference, and action-oriented feedback into one continuous cycle.

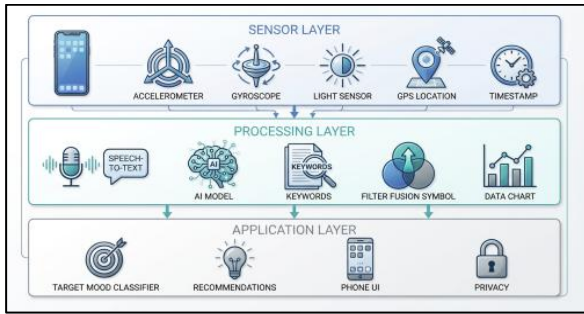


Figure 1: Pocket Therapist System Architecture

Instead of treating journaling, sensing, and recommending as separate features, the Pocket Therapist's design integrates them so that all three functions of Pocket Therapist are interconnected and support each other to enhance each other. The figure illustrates the interaction between the sensor layer, processing layer, and application layer, highlighting how user input and contextual data flow through machine learning inference and recommendation components before returning feedback to the user interface.

#### 3.1 System Overview

The user's initial contact with Pocket Therapist starts with them typing or speaking their reflections, which creates low-friction journaling. The on-device, immediate conversion of voice recordings to text reduces the user's burden of creating journal entries. Additionally, journal entries are interpreted using contextual signals (e.g., when, what & where) so that the user's emotions can be better understood in context rather than as isolated text entries. Therefore, language-based emotion inference and sensor-based features are integrated into a single user state, allowing the Pocket Therapist recommendation engine to provide context-appropriate and timely recommendations for wellness guidance. Ongoing interactions are used to improve the accuracy of inferences in the future, thus making

Pocket Therapist more like an active companion instead of a passive tool for keeping track of journal entries.

### 3.2 Component Interaction

Although the Pocket Therapist platform is technically one system, the level of success achieved by the user depends on the ability of all the system components to work closely together. An example of this is the way in which activity recognition dictates when the wellness recommendations will be sent so that the user's daily routine is not interrupted, whereas the emotion inference process dictates how the recommendations should sound. The user identifies what data can be provided to the system, and if a user hasn't provided data for a specific condition, the recommendation engine depends primarily on the input provided by the user through the journaling process. This requirement creates a balance between providing the user with personalized experiences, while still preserving a level of autonomy with respect to how the system interacts with them. Consequently, these considerations guided the design constraints for Pocket Therapist discussed in the following section.

#### Application Control Flow & Sensor Fusion Pipeline:

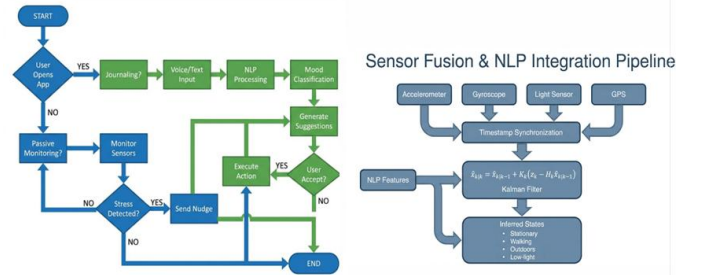


Figure 2: PocketTherapist control flow pipeline

## 4. Design Constraints and Ethical Considerations

### 4.1 Technical Constraints

Due to the limited capabilities of mobile devices to compute, Pocket Therapist incorporated lightweight emotion inference and context models for its mobile application. Users will experience ongoing, positive emotional interchange with Pocket Therapist only when they engage in active journaling, at which time intensive inference and processing may occur. Energy consumption will be mitigated by utilizing adaptive sensing that does not require continuous background operation, activating context inference as opportunities arise. As network conditions vary, the design was adapted to have local first architecture, giving the user the ability to experience the application's primary functions even when offline or online.

### 4.2 Ethical and Societal Considerations

As expected from a product dealing with sensitive information such as journal entries and sensor data, ethical design considerations were paramount. The user has control of their data through explicit,

transparent permission requests for accessing data, and all features remain operational when the user does not permit the use of their data. Additionally, unless the user allows their data to be processed on the server, all data will only be processed and stored locally, with appropriate communication provided regarding any server-side processing of user's journal entries. The design was further refined to include explicit deletion semantics to ensure the user has continuing control over their data. By making the choices, Pocket Therapist's foundational design principles are built around privacy, autonomy, and trust.

## 5. Detailed Design

This section describes the most consequential design decisions, focusing on why specific approaches were chosen and how they address the constraints and goals established earlier.

### 5.1 Emotion and Language Understanding

One of the biggest obstacles in analyzing users' mood from their journal is capturing emotion by interpreting their journal entries. Despite the rise of large language models' generative capabilities being exciting, Pocket Therapist focuses on classifying emotions for both reliability and interpretability. In terms of mental health, it is hard to restrict and clarify the usage of generative outputs; if these models provide incorrect or overly positive responses, it could damage the trust users have in mental health professionals. With emotion classification, mental health professionals can use structured reasoning to help understand users emotionally; however, they encourage users to provide feedback on their feelings through curated recommendations. Instead of simulating conversation, the language processing pipeline provides a way for the user to find out what types of emotional tones are present in the journal, indications of stress, and common themes in the user's journal entries. To do so, natural language models that are computationally lightweight will be used to analyze journal entry text to assign a score for the probable emotional categories and associated confidence of each score, while balancing computational efficiency with accuracy for on-device inference. Emotional states can be assigned to users as probability signals, not as a diagnosis; therefore, users will have less likelihood of overreacting to individual entries. By aggregating emotional signals over time, while still allowing for uncertainty, the Pocket Therapist system places greater weight on the user's emotional trends while continuing to support rather than prescribe to them.

### 5.2 Sensor Fusion and Activity Recognition

Due to device placement, behavioral variability and other sources of noise, there are many challenges involved in inferring context from mobile phone sensors. Individual signals, such as accelerometer data and step counts, can often be very ambiguous when considered in isolation; therefore, Pocket Therapist uses sensor fusion based on multiple signals to infer activity more accurately and robustly. A Kalman filter is used to combine and smooth motion-related signals

over time, allowing the system to have a continuously updated estimate of each user's activity state while also compensating for transient noise. The impact of hardware on activity recognition adds further complexity to the inference process since each device has varying sensor sensitivity and running background processes differently. Consequently, activity recognition derives its results from observing relative changes over time, rather than from reliance on absolute thresholds, thus enabling the system to be generalized across multiple devices without requiring calibration on each one. Through this type of design, reliability is enhanced, thereby ensuring that users' behavior will remain consistent while using heterogeneous smartphones.

### 5.3 Recommendation Engine

The recommendation engine turns insights gained from emotions and other context into real-world actions and suggested interventions that are both timely and respectful. The approach is to use the interplay between an individual's recent history of activities, the current emotional tone, and the timeframe of the interaction to inform the recommendations and contextualize them within the user's experience. For example, during times of inactivity when emotional states are elevated (e.g., stressed), calming exercises may be recommended; however, similar recommendations may not necessarily occur during active times. Additionally, a core principle of transparency and understanding (or "explainability") is incorporated into all recommendations by including a short narrative or justification connecting them to the most recently entered journal entries or corresponding contextual patterns. This practice fosters a sense of transparency and trust in the product/service. Another major design feature of the engine is a temporal and diversity constraint which helps prevent the user from feeling overwhelmed by repetitive or similar feedback. Interventions are also throttled and rotated in terms of frequency to promote novelty in relation to ongoing emotional state patterns. This design feature keeps users continuing to engage with the service/product and helps to eliminate excessive cognitive load.

### 5.4 User Interface Design

The user interface is designed to make system complexity largely invisible while keeping interaction simple and low effort. One of the main challenges was creating a voice journaling experience that feels natural rather than intrusive. Although voice input reduces friction, it raises concerns around transcription accuracy and user comfort. To address this, the interface allows easy switching between voice and text, with clear options to review, edit, and confirm entries. Permission timing was another important design consideration. Requesting access too early can feel overwhelming, while delaying permissions can interrupt core functionality. Pocket Therapist addresses this by requesting permissions only when they become relevant, aligning system requests with user intent. Recommendations are presented in a layered manner, appearing concise by default and

expandable when users want more detail. Throughout the interface, visual hierarchy and spacing are used intentionally to guide attention and reduce cognitive load. Together, these choices emphasize clarity, robustness, and user agency rather than heavy automation.

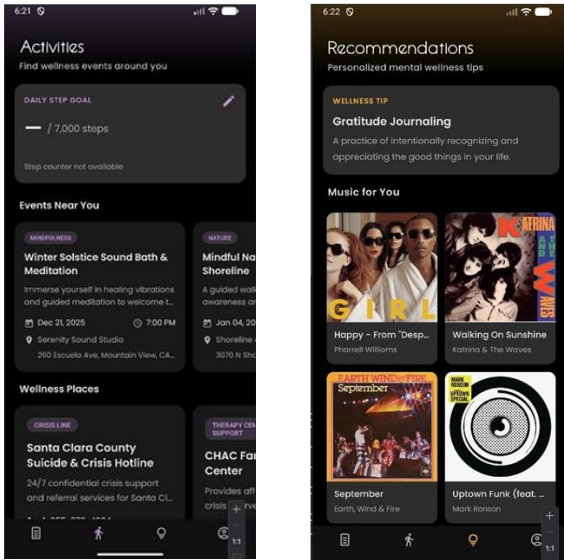


Figure 3: The Activities and Recommendation UI

## 6. Implementation

Implementing Pocket Therapist involved translating a system-level design into a robust Android application that operates reliably across devices, permissions, and real-world usage. Although individual features were simple in isolation, complexity arose from coordinating asynchronous components, managing Android lifecycle constraints, and maintaining stable sensing and inference over time.

### 6.1 Interesting Implementation Aspects

Orchestration of asynchronous pipelines that integrate user input, sensor data, and machine learning inference; Voice Journaling required orchestration of asynchronous pipelines between Speech Recognition Callbacks, Local Storage, and downstream Emotion Inference Operations without halting the User Interface (UI). The Voice Journaling component involved separate processing modules executed in parallel threads along with lifecycle aware components which were used to ensure that partial or unfinished results would not persist across either configuration changes (like orientation change) or interrupted sessions (like when the app is sent to the background). Since much of the UI relied on sensors and processing that could occur in the background, Lifecycle Management was essential. All Activity Tracking and Context Inference must be consistently persistent across both App Restart and Device Reboot while adhering to Android's limits on Background Execution. Therefore, Sensing Components had to be implemented as Foreground Services that presented themselves to users; it was necessary to implement State Restoration Mechanisms to ensure that data was not lost due to App Pause or Retrieve events. Additionally,

since multiple sensors needed to stay temporally aligned, even when the App was inactive, Sensor Persistence was also a challenge. To address this, summarized sensor features were stored, instead of raw streams, allowing efficient reconstruction of context while dramatically reducing the amount of storage consumed and power required. Assuming these design considerations, the app will be a long term, or consistently available, "Companion" App as opposed to merely a brief Foreground App.

### 6.2 Major Challenges and Resolutions

Several challenges were faced during testing and in the final demonstration. For example, when interrupted or rapidly repeated audio recordings, speech recognition would sometimes return error codes. This required tracing through the lifecycle callbacks and resetting the recognizer state between sessions; the final solution handled this by falling back on text input so as not to interfere with user flow. The behavior of the step counter varied across devices, with some devices reporting increments even when the user was not in motion. To resolve this, our team treated the step count as a complement to other motion sensors within the fusion pipeline and weighted it relative to those sensors. This led to more robust results while reducing the hardware sensitivity of the delivery system. Bug fixes related to user permissions showed the need for users to give explicit consent regarding access to their data and information. Early versions of the system assumed that permission would always be granted; however, when permissions were either not provided or revoked, the system failed to function. Thus, the team implemented checks to ensure that the user was aware of their permission status prior to any operations that required data dependency on that permission. Furthermore, the NAVIGATION FLOWS were restructured so that data-dependent actions would only be executed after the user granted their consent. A verification system was also placed to confirm that an HTTPS Endpoint was contacted prior to making any external requests. All these fixes contributed to increased reliability and durability, while at the same time, increasing user trust in the system.

## 7. Evaluation

### 7.1 Testing Approach

Testing followed a layered strategy aligned with the system architecture. Functional testing validated end-to-end workflows such as journaling, emotion inference, and recommendation delivery, with particular focus on foreground and background lifecycle transitions. Stress testing involved repeated journaling sessions, extended background sensing, and rapid screen navigation to identify memory leaks, race conditions, and state inconsistencies in asynchronous pipelines. Device variability testing was conducted across multiple Pixel devices to account for differences in sensor behavior and inference latency, informing refinements to sensor fusion and lifecycle handling.

7.2 Measurements

Latency and resource utilization were evaluated quantitatively. Both local and cloud/server-based Emotion Classification systems were compared based on inference latency, with the local solution producing a consistently lower average and more repeatable response times as compared to the remote option. Memory consumption was also analyzed during extended journaling sessions with continuous sensing at high rates, demonstrating steady memory utilization patterns as the summarized sensor features were utilized as opposed to raw data streams. Furthermore, the performance differences perceived among devices were determined to be more the result of differences in hardware rather than limitations in architecture.

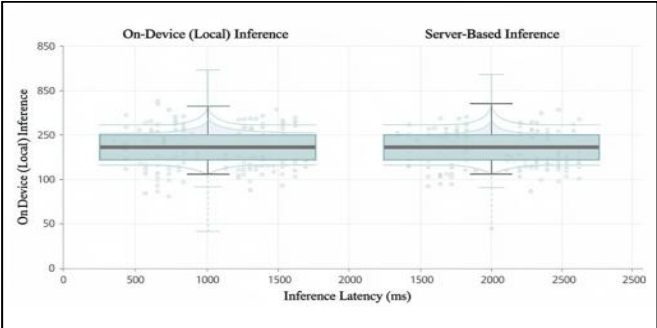


Figure 4: Local vs. Server-Based Emotion Inference Latency Comparison

Figure 4 presents a comparison of local versus server-based inference latency, highlighting the reduced variance and improved responsiveness achieved through local processing.

7.3 User Study Results

A user study with 12 participants evaluated usability, perceived usefulness, and overall experience. Full survey distributions are included at the end of the report; key findings are summarized here. Participants consistently rated Pocket Therapist as easy to use, quick to learn, and well-integrated, with high confidence in using the system without technical support. Perceived complexity and inconsistency were low. Low friction was a central strength. Voice journaling reduced effort and was successfully used by most participants on their first attempt. Context-aware recommendations were generally well received, particularly when brief explanations were provided, though some users desired clearer confirmation before external actions. Pre-study responses indicated that time burden and privacy concerns commonly led to abandonment of prior journaling tools, suggesting that Pocket Therapist addresses major adoption barriers.

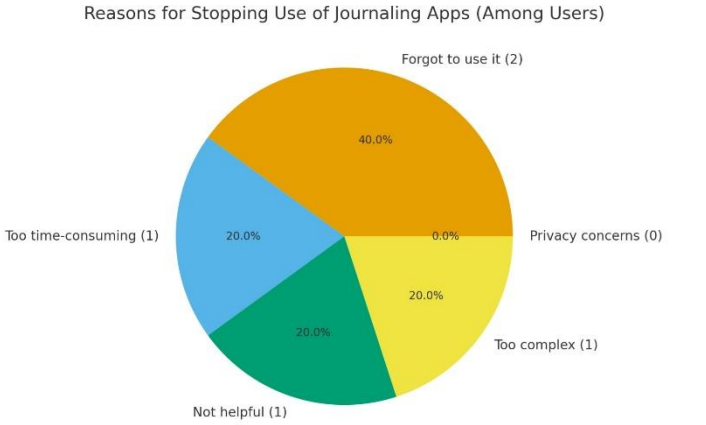


Figure 5: Reasons users stated for stopping the use of journaling apps

7.4 Analysis of Results

Overall, results show that Pocket Therapist reduces journaling friction while providing timely, context-aware support. Voice input, passive sensing, and on-device inference enabled low cognitive load and reliable performance across devices. Opportunities for improvement include longer-term personalization and improved calibration of activity inference. User feedback also emphasized the importance of transparency and explicit consent. Collectively, the findings support the feasibility of balancing intelligent support with usability and trust in mobile wellness systems.

| Use frequently | Unnecessarily complex | Easy to use | Need tech support | Functions integrated | Too inconsistent | Learn quickly | Cumbersome | Confident using | Needed to learn a lot |
|----------------|-----------------------|-------------|-------------------|----------------------|------------------|---------------|------------|-----------------|-----------------------|
| 4              | 2                     | 4           | 1                 | 4                    | 1                | 3             | 1          | 4               | 1                     |
| 5              | 1                     | 5           | 1                 | 3                    | 1                | 5             | 1          | 4               | 1                     |
| 5              | 1                     | 3           | 2                 | 5                    | 2                | 5             | 2          | 3               | 1                     |
| 4              | 1                     | 4           | 1                 | 5                    | 1                | 5             | 2          | 3               | 1                     |
| 3              | 1                     | 4           | 1                 | 5                    | 1                | 5             | 3          | 3               | 3                     |
| 3              | 2                     | 3           | 1                 | 5                    | 1                | 3             | 1          | 4               | 2                     |
| 3              | 1                     | 4           | 1                 | 4                    | 1                | 4             | 1          | 4               | 2                     |
| 5              | 1                     | 3           | 3                 | 5                    | 2                | 3             | 2          | 4               | 3                     |
| 4              | 2                     | 3           | 1                 | 3                    | 1                | 5             | 2          | 5               | 2                     |
| 5              | 1                     | 5           | 2                 | 3                    | 3                | 4             | 2          | 4               | 1                     |
| 3              | 1                     | 5           | 1                 | 3                    | 2                | 4             | 2          | 4               | 3                     |
| 5              | 1                     | 5           | 1                 | 4                    | 1                | 3             | 1          | 5               | 1                     |

Figure 6: Post-Study Usability Survey Responses Across Participants

8. Lessons Learned, Contributions, and Insights

8.1 Team and Process

The team adopted a Scrum-like, iterative workflow, coordinating daily and integrating frequently. There was no formal Scrum master, but this was accommodated through short development cycles and regular check-ins that allowed for quick feedback and modifications to work products developed by team members.



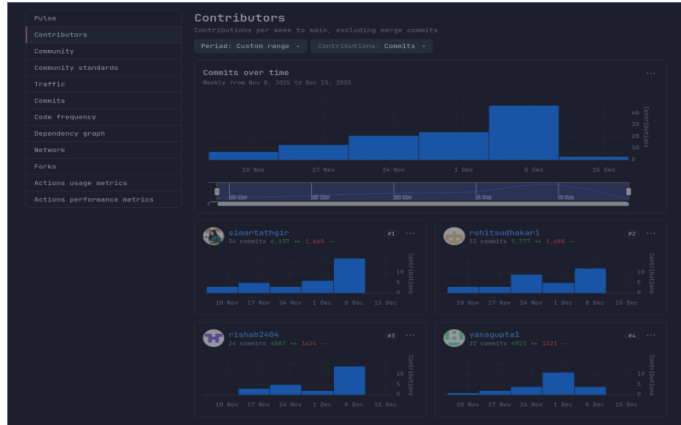


Figure 6: Graph of github commits

Version control of code was managed with GitHub, using feature branches and pull requests to allow for the development and review of features to occur in parallel to one another. As the size and complexity of the system increased, practices associated with committing code to GitHub became an important tool for tracking development and resolving development regressions. Figure 4 provides an illustration of GitHub commits over time, providing an overview of the contribution of each team member and changes in areas of responsibility as the team integrated members as part of an iterative integration process.

## 8.2 Contributions

The contributions were divided but did overlap. One contributor developed for Android operating system developing its life cycle management, sensor integrations as well as user-interface implementation. Another contributor created the machine learning and inference pipeline, working on the emotion classification and contextual reasoning components and implemented machine learning for this project. A third contributor focused on the integration of all components creating a coordinated testing and user study. Through shared debugging and integration accomplished within every contributor's effort; shared understanding between the contributors is reduced in bottleneck experiences.

## 8.3 Retrospective Insights

One important lesson learned through this project is the early integration of the entire system level. When components are treated as individual components, performance and lifecycle issues are discovered later in the process. The result of this project also emphasized the trade-offs between the use of continuous sensing on mobile devices. This has reinforced the idea that the use of sensible instrumentation and transparency is necessary in mobile wellness systems. Therefore, the project has established that there needs to be a careful balance between the three main factors of privacy, usability and technical limitations in mobile wellness systems.

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